EXPERIMENT - 11(a)

29/9/26 STATIC ROUTING

AIM:

To simulate and configure static routing with primary and backup routes using CISCO Packet tracer and verify connectivity between different networks.

PROCEDURE:

1) Open CPT and create given topology with Routers(0-2), switches and PCs.
2) Configure IP address on all devices
3) Open the CLI of Router 0 and enter configuration mode
4) Configure static router for networks 30.0.0.0/8 and 50.0.0.0/8 using Router 1 - primary route Router 2 - backup route
5) Configurate a seperate host route for router 30.0.0.10/32.
6) On Router1, configure static routes for network 10.0.0.0/8 and 40.0.0.0/8
7) On Router2, configure static routes 10.0.0.0/8 and 30.0.0.0/8
8) Use the command: show ip route static to verify the configured static routes
9) Use the tracert command from a PC to verify the path taken by packets to the destination network.
10) Disconnect the primary link between R0 and R1 and use tracert again to verify that the backup route became active
11) Static routes can be removed using: no IP route
12) Verify the route table again using show ip route static.

RESULT:

Thus the routing with primary and backup routes was successfully configured and simulated using the cisco pkt Tracer

EXPERIMENT - 11(b)

29/9/26. RIP (Routing Information Protocol)

AIM: To simulate and configure RIP (Routing Inform. Protocol) using Cisco Packet Tracer and verify dynamic routing between different networks.

PROCEDURE:

1) Open Cisco Packet Tracer and create the topology with Router(0-2), PC0 and PC1.
2) Assign IP addresses to PCs and routers interfaces
3) Configure the FastEthernet and serial interfaces of all routers using the assigned IP addresses
4) Use no shutdown to activate all router
5) For DCE serial interfaces, configure the required clock rate and bandwidth
6) Configure RIP on R0 using:-

```
router rip
network 10.0.0.0
network 192.168.1.252
```

7) " " R1 " :-

```
router rip
network 192.168.1.244
network 192.168.1.248
```

8) " " R2 " :-

```
router rip
network 192.168.1.252
network 192.168.1.249
```

9) Allow to RIP to exchange Routing information
10) Verify connectivity (PC0 to PC1), using ping command
11) Use the tracert command to observe the path followed by packets between the n/w
12) Disconnect one of the available routes and use tracert again to verify that RIP automatically.

RESULT:

Thus RIP was successfully configured and simulated using CPT